

Reviving the Lost Wilderness: Measuring the Educational Value of Immersive Technologies in Prehistoric Wildlife Exploration

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Final Project Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research investigates whether augmented reality (AR) museum tours delivered through visitors' mobile phones can improve wayfinding, engagement, and learning outcomes compared to traditional printed label tours. Many museums in Sri Lanka still rely on static interpretation methods that limit visitor interaction and engagement, particularly among younger audiences. To address this, a mobile AR based Fossil Tour application was developed using Unity and AR Foundation, incorporating AR navigation, interactive 3D models, and a Virtual AR Guide that provides on-demand information. A comparative experimental study was conducted at the Colombo National Museum, where participants first experienced a traditional tour and then the AR based tour, completing identical quizzes in both conditions. Results showed a significant improvement in learning outcomes and engagement when using the AR system, with higher average scores and increased interaction levels. The findings indicate that AR enhances immersive, self-paced learning while improving navigation, demonstrating that mobile AR solutions offer a cost effective and scalable approach for enhancing museum experiences and supporting informal learning.

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LIST OF ABBREVIATIONS

Abbreviations	Description
AR	Augmented Reality
VR	Virtual Reality
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
UI	User Interface
NLP	Natural Language Processing
RAG	Retrieval Augmented Generation
DSR	Design Science Research
ESP32	Microcontroller used for BLE beacon implementation
RSSI	Received Signal Strength Indicator
IoT	Internet of Things
3D	Three Dimensional
GPT	Generative Pre trained Transformer

INTRODUCTION

Background and Literature Survey

Museums are widely recognized as important informal learning environments that facilitate knowledge acquisition outside traditional classroom settings. Unlike formal education, museum learning is typically self-directed, voluntary, and driven by individual curiosity. This aligns with constructivist learning theory, where learners actively construct knowledge through interaction with their surroundings rather than passively receiving information. In museum contexts, learning occurs through observation, interpretation, and engagement with physical artifacts. However, traditional museum practices continue to rely heavily on static exhibits supported by printed labels, which present information in condensed textual formats. This approach often requires visitors to interpret complex concepts independently, limiting engagement and comprehension, particularly among younger audiences or those with limited prior knowledge [2].

These limitations are further pronounced in Sri Lankan museums, where the integration of digital technologies remains minimal. Fossil exhibits, for instance, typically consist of skeletal remains or fragmented artifacts that require significant imagination to interpret. Without visual reconstruction or contextual storytelling, visitors may struggle to understand the significance of these exhibits, resulting in passive experiences with minimal long-term learning outcomes [6]. At the same time, the rapid advancement of digital technology has transformed user expectations. Modern users, especially younger generations, are accustomed to interactive, visually rich, and responsive digital environments such as mobile applications and gaming platforms. These environments offer immediate feedback, user control, and immersive interaction, creating a clear gap between user expectations and traditional museum experiences [3].

Augmented Reality (AR) has emerged as a transformative technology capable of bridging this gap by overlaying digital content onto the physical environment in real time. In museum settings, AR enables the visualization of reconstructed artifacts, animation of extinct organisms, and delivery of contextual information through

interactive storytelling. Unlike Virtual Reality (VR), which isolates users in fully digital environments, AR enhances the existing physical context while preserving real-world interaction. Mobile AR solutions are particularly advantageous due to their accessibility, cost-effectiveness, and scalability, as they utilize widely available smartphones without requiring specialized hardware [2].

Numerous studies have demonstrated the effectiveness of AR in enhancing learning outcomes. Wen and Ma [1] highlight that AR systems integrated with intelligent technologies can significantly improve user engagement and personalized learning experiences. Similarly, Boboc et al. [2] emphasize that AR enhances contextual understanding by allowing users to visualize abstract concepts within real-world environments. Cheng et al. [3] further identify psychological factors such as immersion, motivation, and enjoyment as key contributors to improved learning performance in AR based systems. From a cognitive perspective, AR supports dual coding theory, which suggests that combining visual and verbal information enhances memory retention and recall [4].

Despite these advantages, both traditional and AR based systems must address challenges related to cognitive load. Cognitive Load Theory suggests that learning effectiveness depends on the amount of mental effort required to process information. Traditional label based systems often impose high cognitive load due to dense textual information and lack of visual support. Conversely, poorly designed AR systems can also overwhelm users through excessive information, continuous narration, or complex interfaces. Therefore, effective AR systems must balance interactivity with simplicity, ensuring that users can control the flow of information and engage at their own pace.

Another critical aspect of museum experience is wayfinding, which refers to how visitors navigate physical spaces to locate exhibits. Traditional wayfinding methods such as static maps and signboards often require interpretation and provide no real-time feedback, leading to confusion and inefficient exploration. Research in Human-Computer Interaction suggests that context-aware navigation systems significantly improve user experience. Bluetooth Low Energy (BLE) technology offers a practical

solution for indoor navigation by enabling proximity based positioning using signal strength measurements. BLE beacons can guide users in real time through audio or haptic feedback, reducing reliance on visual maps and enhancing navigation efficiency.

In addition to navigation, the role of guided interaction is essential in enhancing museum learning experiences. While human guides provide personalized explanations, they are resource intensive and not always available. AI driven Virtual Guides offer a scalable alternative by enabling natural language interaction, adaptive explanations, and user driven learning. By integrating speech recognition, natural language processing, and text to speech technologies, virtual guides can create conversational and inquiry based learning environments that closely resemble human interaction.

Furthermore, gamification techniques have been widely used to increase user engagement in educational environments. The incorporation of game elements such as quizzes, rewards, and progression systems encourages active participation and reinforces learning. Studies indicate that gamified AR experiences significantly improve motivation, engagement, and knowledge retention compared to traditional methods [7].

Although existing AR based museum systems demonstrate strong visualization capabilities, many lack integration across key components such as navigation, interaction, learning evaluation, and analytics. Additionally, several systems rely on expensive hardware or infrastructure, limiting their practical deployment in developing regions. This research addresses these limitations by proposing an integrated solution that combines mobile AR visualization, BLE based navigation, AI driven interaction, gamification, and real time analytics within a single system. By doing so, it aims to provide a cost effective, scalable, and user centered approach to enhancing museum experiences and supporting informal learning.

Research Gap

Although extensive research has explored the application of Augmented Reality (AR) in museum environments, a critical review of existing literature reveals several unresolved limitations that highlight the need for a more integrated and user-centered solution. One of the most significant gaps is the limited focus on wayfinding and spatial navigation. Most AR applications primarily enhance individual exhibits through digital overlays such as 3D models and textual information, while largely neglecting the challenge of guiding visitors through the overall museum space. Traditional navigation methods, including static maps and signboards, require manual interpretation and often lead to spatial confusion, especially in complex layouts. Existing digital solutions are typically limited to map-based interfaces and lack real-time, context aware guidance. This indicates the absence of systems that seamlessly integrate navigation with exhibit interaction, which is essential for improving visitor experience in real world environments.

Another critical limitation is the over reliance on passive AR content delivery mechanisms. Many current AR systems utilize automatic audio narration or pre defined content sequences, which reduce user autonomy and limit interactive engagement. Such passive approaches may lead to cognitive overload, reduced attention span, and limited personalization, as information is presented regardless of user interest or readiness. Effective learning environments, however, require active participation and user control over the pace and depth of information. Despite this, most existing systems do not support inquiry based interaction or allow users to request information dynamically, creating a disconnect between technological capabilities and natural visitor behavior in museums.

Furthermore, there is a notable lack of empirical comparison between AR based systems and traditional printed label museum tours. While many studies highlight the benefits of AR in terms of engagement and user satisfaction, they often fail to provide rigorous comparative evaluations that assess whether AR significantly outperforms conventional methods. Most evaluations focus on subjective metrics such as user perception or system usability,

rather than measurable outcomes such as learning effectiveness and navigation efficiency. This limits the ability to draw meaningful conclusions about the true impact of AR technologies in museum contexts.

In addition, many existing AR implementations rely on high cost hardware such as head mounted displays or custom installations, which restrict scalability and practical deployment, particularly in developing countries. Although mobile AR offers a more accessible alternative, there is limited research on designing low cost, scalable systems that can be easily integrated into existing museum environments without requiring major infrastructure changes. This creates a gap in developing practical solutions that are both technologically effective and economically feasible.

Another limitation in current research is the lack of holistic evaluation frameworks. Museum experiences are inherently multi dimensional, involving navigation, interaction, learning, and emotional engagement. However, most studies evaluate AR systems based on a single dimension, such as usability or engagement, without considering the overall visitor experience. This fragmented approach prevents a comprehensive understanding of system effectiveness and limits the ability to optimize design across multiple factors.

Finally, despite the rapid advancement of artificial intelligence, its integration within AR based museum systems remains underutilized. Most existing applications rely on static or pre scripted content, without leveraging AI for dynamic interaction or personalization.

AI driven systems have the potential to enable adaptive content delivery, conversational interaction, and real time user support, yet few studies explore these capabilities within AR environments. This research addresses these identified gaps by proposing an integrated system that combines mobile AR visualization, BLE based navigation, AI driven virtual guidance, gamification, and comprehensive analytics. By doing so, it aims to provide a scalable, cost effective, and interactive solution that enhances wayfinding, engagement, and learning outcomes in museum environments.

Do AR museum tours on visitors' phones improve wayfinding, engagement, and learning versus a printed label tour?

Traditional museum systems that rely on printed labels and static displays present several limitations in supporting effective visitor experiences. These systems provide minimal assistance for navigation, often requiring visitors to interpret maps or signage independently. As a result, visitors may experience spatial confusion, miss important exhibits, or spend excessive time locating points of interest. Such limitations negatively affect the overall efficiency of museum exploration and reduce the likelihood of meaningful engagement with exhibits [2].

In addition, traditional exhibits offer limited interactivity, which significantly reduces visitor engagement. Visitors typically spend only a short amount of time at each exhibit, resulting in shallow cognitive processing and limited knowledge retention. The absence of interactive elements and personalized content further restricts the ability to accommodate diverse learning preferences, particularly for younger audiences who are more accustomed to dynamic and visually rich digital environments [3]. Studies have shown that passive learning methods, such as reading static text, are less effective in promoting long term memory retention compared to interactive and multimodal learning approaches [4].

Augmented Reality (AR) technologies have emerged as a promising solution to enhance museum experiences by providing interactive visualization and contextual information. AR enables the integration of digital content with physical exhibits, allowing users to engage with reconstructed artifacts, animations, and guided explanations in real time.

Prior research indicates that AR based systems can improve user engagement, motivation, and learning outcomes in educational environments [1][2]. However, despite these advantages, existing AR implementations in museums exhibit several limitations.

Many current AR systems primarily focus on visualization and lack integrated support for navigation. As a result, users may still face challenges in locating exhibits efficiently within large or complex museum environments. Furthermore, several AR applications rely on passive content delivery mechanisms, such as automatic audio narration or predefined information sequences, which limit user control and may increase cognitive load. Such approaches do not fully support inquiry based learning, where users actively explore content based on their interests and curiosity [5].

Another critical limitation is the lack of empirical evaluation comparing AR based museum systems with traditional printed label methods. While existing studies often report improvements in user satisfaction and engagement, they rarely provide comprehensive, data driven comparisons across multiple dimensions such as wayfinding efficiency, learning outcomes, and interaction behavior. This lack of comparative analysis makes it difficult to determine the true effectiveness of AR technologies in real world museum contexts [7].

Moreover, many advanced AR solutions depend on expensive hardware or complex infrastructure, limiting their scalability and practical implementation, especially in developing countries. There is a need for a cost effective and accessible solution that leverages widely available technologies such as smartphones, while still delivering enhanced user experiences.

Therefore, the central research problem can be articulated as follows:

How can a mobile augmented reality museum system, enhanced with BLE based proximity navigation and an AI driven Virtual AR Guide, be designed and evaluated to improve wayfinding, visitor engagement, and learning outcomes compared to traditional printed label museum tours?

Addressing this research problem is particularly important in the context of Sri Lankan museums, where resource constraints demand low cost and scalable technological solutions. By integrating mobile AR, BLE based navigation, and AI driven interaction into a unified framework, this research aims to bridge the gap between technological innovation and practical implementation. Furthermore, the study seeks to provide empirical evidence through real world evaluation, ensuring that the proposed system not only enhances user experience but also delivers measurable improvements in learning and engagement.

Research Objectives

Main Objective

The primary objective of this research is:

To design, develop, and evaluate a mobile augmented reality (AR) museum tour system enhanced with BLE based proximity navigation and an AI driven Virtual AR Guide, with the aim of improving wayfinding, visitor engagement, and learning outcomes compared to traditional printed-label museum tours.

This objective reflects a multi-dimensional research goal, combining system development with empirical evaluation. Unlike purely technical projects, this research does not focus solely on building an application but also on assessing its effectiveness in a real-world context.

The main objective is structured around three core dimensions:

- Wayfinding – Improving how visitors navigate within the museum
- Engagement – Increasing user interaction and time spent with exhibits
- Learning – Enhancing knowledge acquisition and retention

By addressing these three aspects simultaneously, the research aims to provide a holistic improvement to the museum experience, rather than focusing on a single isolated factor.

Specific Objectives

To achieve the main objective, a set of interrelated specific objectives has been defined. Each objective targets a particular component of the system and contributes to the overall research goal.

To Develop an AR Visualization System

This objective focuses on designing and implementing a mobile AR system capable of overlaying interactive 3D models onto real world museum exhibits.

Traditional fossil displays often present incomplete or abstract representations of prehistoric life. Visitors are required to interpret skeletal remains or fragments without visual context, which limits understanding. The AR visualization system addresses this limitation by reconstructing fossils into animated 3D models that represent the living organism.

The system is developed using Unity and AR Foundation, enabling cross-platform compatibility and real-time rendering. Users can interact with the 3D models through gestures such as scaling, rotation, and positioning, allowing for detailed examination.

This objective contributes to:

- Enhancing visual comprehension
- Supporting experiential learning
- Bridging the gap between abstract data and realistic representation

To Implement BLE Based Proximity Navigation

This objective aims to design a navigation system that assists users in locating exhibits within the museum using Bluetooth Low Energy (BLE) technology.

Traditional navigation methods rely on static maps or signage, which require interpretation and can lead to confusion. In contrast, BLE based navigation provides real time, context aware guidance by detecting the proximity of the user to specific exhibits. ESP32 BLE beacons are deployed at exhibit locations, continuously broadcasting signals. The mobile application scans these signals and calculates proximity using RSSI values. Based on this data, the system provides feedback such as audio cues or visual indicators to guide users toward their target. This objective,

contributes to:

- Reducing navigation difficulty and cognitive load
- Improving spatial awareness
- Enhancing user flow within the museum environment

To Design an AI Driven Virtual AR Guide

This objective involves developing an intelligent Virtual Guide that provides interactive, user controlled information delivery.

Unlike traditional systems that rely on automatic narration, the Virtual Guide enables users to actively request information through voice interaction. The system uses speech recognition, natural language processing, and text-to-speech technologies to create a conversational interface.

The Virtual Guide is designed to:

- Answer user questions
- Provide explanations about exhibits
- Adapt responses based on user age or knowledge level

This approach aligns with inquiry based learning principles, where users engage more deeply by asking questions and exploring content at their own pace. This objective contributes to:

- Increasing user autonomy
- Supporting personalized learning experiences
- Reducing information overload

To Integrate Gamification Mechanisms

This objective focuses on incorporating game based elements into the system to enhance motivation and engagement.

Gamification is implemented through features such as:

- Multiple choice quizzes
- Reward systems (e.g., coins or points)
- Progression tracking and unlocking mechanisms

These elements encourage users to actively participate in the learning process rather than passively consuming information. Quizzes reinforce knowledge retention by requiring users to recall and apply what they have learned.

This objective contributes to:

- Increasing time spent at exhibits
- Encouraging active participation

To Evaluate User Engagement

This objective aims to assess how effectively the system captures and maintains user attention.

Engagement is measured using both quantitative and qualitative metrics, including:

- Time spent at exhibits
- Number of interactions (e.g., model manipulation, guide queries)
- Frequency of feature usage
- User feedback through surveys

To Measure Learning Outcomes

This objective focuses on evaluating the educational impact of the system. Learning outcomes are assessed using:

- Pre and post experience quizzes
- Knowledge retention tests
- User self assessment surveys

The goal is to determine whether the AR system improves:

- Knowledge acquisition
- Conceptual understanding
- Long term retention

This objective is critical because it moves beyond user satisfaction and examines the actual educational value of the system.

To Compare with Traditional Museum Tours

This objective involves conducting a comparative study between the AR based system and traditional printed label tours.

Participants are divided into groups:

- One group uses the AR system
- Another group experiences the traditional method

The comparison focuses on:

- Navigation efficiency
- Engagement levels
- Learning outcomes

This objective provides empirical evidence to determine whether the proposed system offers measurable improvements over existing methods.

To Assess System Feasibility

This objective evaluates the practicality of implementing the system in real world museum environments.

Feasibility is assessed across multiple dimensions:

- Technical feasibility – System performance and reliability
- Financial feasibility – Cost of deployment and maintenance
- Operational feasibility – Ease of integration into museum workflows
- User feasibility – Accessibility and usability for diverse audiences

This ensures that the system is not only effective but also realistic and scalable.

METHODOLOGY

Methodology Introduction

This chapter presents a comprehensive explanation of the methodological approach adopted for the design, development, implementation, and evaluation of the proposed Augmented Reality (AR) Fossil Interpretation System with an AI driven Virtual Guide and BLE based navigation. The methodology is structured to ensure both technical rigor and academic validity, addressing the research problem of improving museum wayfinding, engagement, and learning outcomes compared to traditional printed-label systems.

The proposed system integrates multiple technological domains including mobile augmented reality, artificial intelligence, Bluetooth Low Energy (BLE) based indoor positioning, gamification, and data analytics. These components are combined to create an interactive, user centered learning environment that transforms passive museum visits into engaging digital experiences.

Research Methodology Framework

Design Science Research (DSR) Approach

This research adopts the Design Science Research (DSR) methodology, which focuses on the creation and evaluation of innovative artifacts to solve real world problems.

The process includes:

- Problem Identification
- Objective Definition
- Artifact Design
- Development
- Evaluation

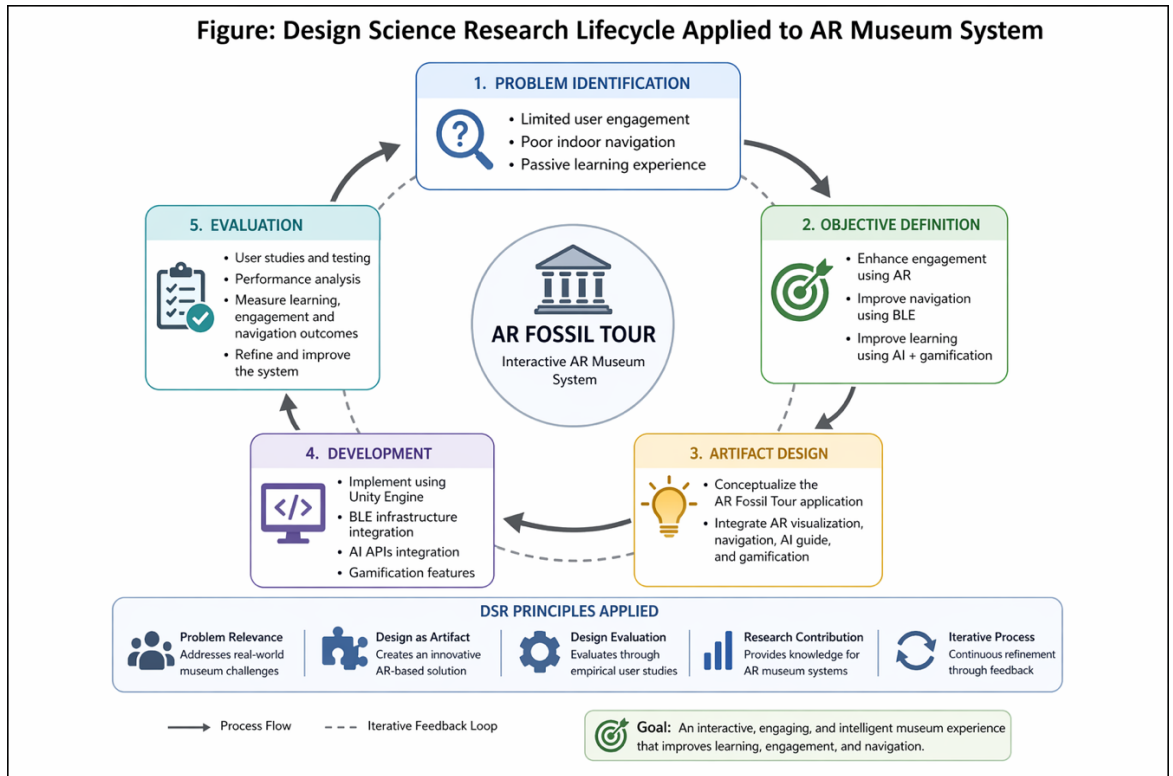


Figure 1: Design Science Research Lifecycle Applied to AR Museum System

System Architecture and Design

The system is designed using a modular layered architecture consisting of:

- Infrastructure Layer
- Service Layer
- Logic Layer
- Presentation Layer

This separation ensures scalability and efficient integration of AR, AI, and BLE technologies.

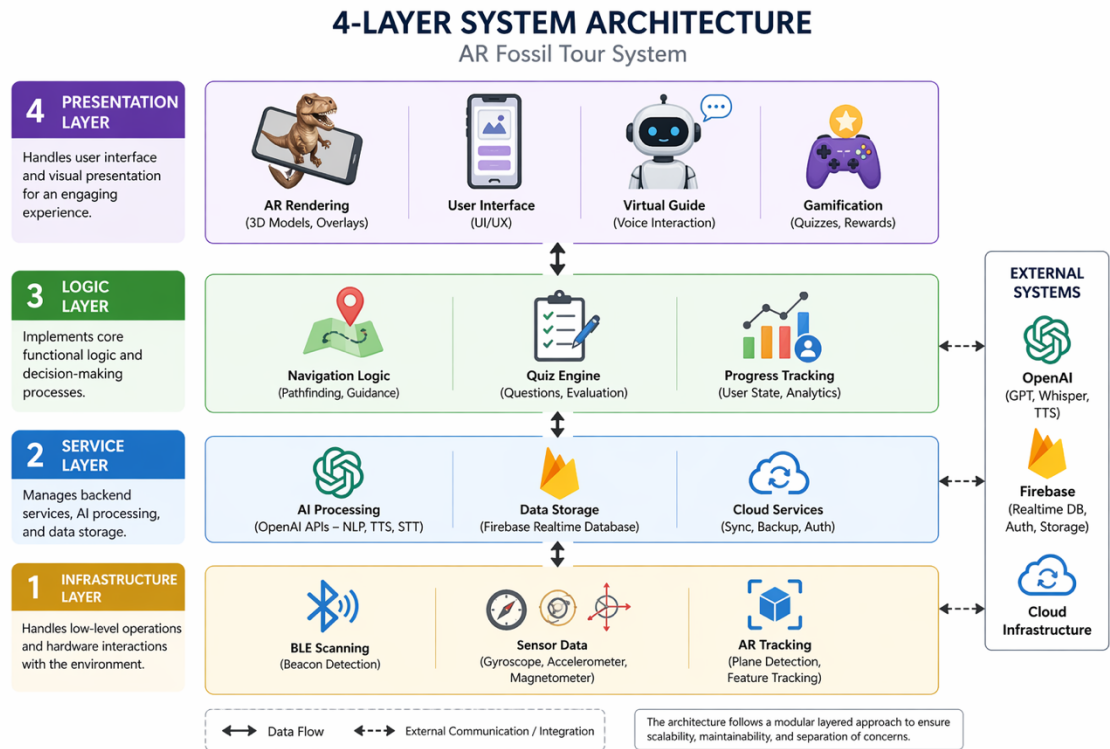


Figure 2: Four Layer System Architecture Diagram

Augmented Reality Module

The AR module enables interactive visualization of fossil exhibits using Unity AR Foundation and ARCore.

Core Functionalities

- Plane detection
- 3D model placement
- Gesture interaction (rotate, scale, inspect)

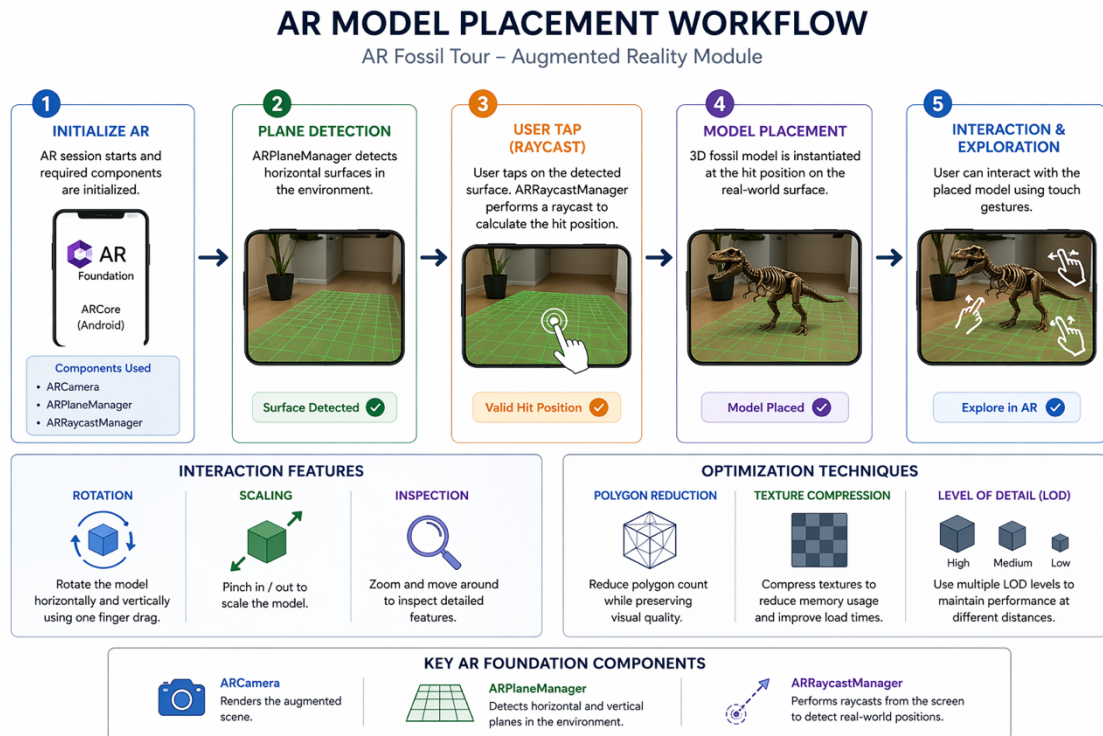


Figure 3: AR Model Placement and Interaction UI

This function is responsible for placing 3D fossil models onto detected surfaces using AR raycasting. It enables users to interact with digital reconstructions in real world space, significantly improving engagement compared to static displays.

BLE Based Navigation System

In this system, ESP32 DevKit modules are configured as BLE beacons to support proximity-based exhibit discovery. Each ESP32 operates in advertiser mode and continuously broadcasts a unique beacon name,

ESP32_Exhibit1, ESP32_Exhibit2, ESP32_Exhibit3 and ESP32_Exhibit4. These identifiers allow the mobile application to recognize which exhibit the user is approaching. The ESP32 does not require a direct connection with the mobile phone; instead, it broadcasts BLE advertising packets, making the system lightweight, low-cost, and suitable for museum deployment .

The mobile application acts as the BLE scanner. It continuously searches for beacon names that match the exhibit pattern and extracts the Received Signal Strength Indicator (RSSI) value from the detected signal. The RSSI value is then used to estimate the user's proximity to the exhibit. When the signal becomes stronger, the system assumes that the user is closer to the target exhibit and increases audio or haptic feedback intensity. In the implemented system, an RSSI threshold of approximately 52dBm is used to identify when the user is close enough to activate the AR content .

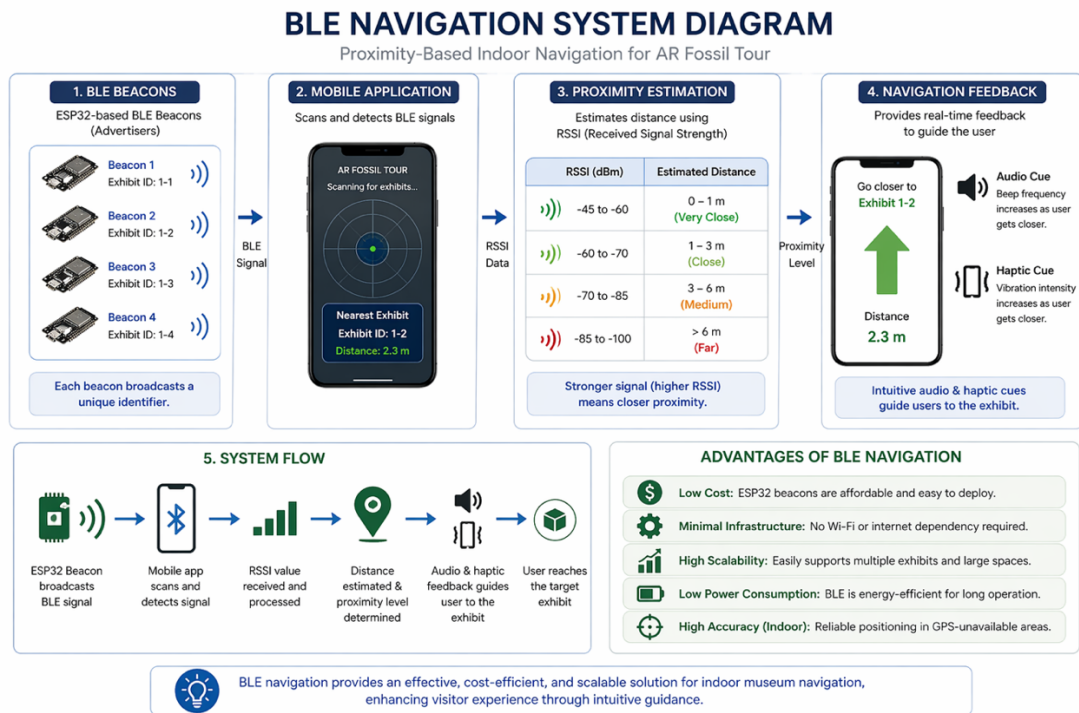


Figure 4: ESP32 BLE Beacon Configuration and Mobile App Detection Flow

```

1  #include <BLEDevice.h>
2  #include <BLEUtils.h>
3  #include <BLEServer.h>
4  // Unique name for each exhibit (e.g., ESP32_Exhibit1)
5  #define BEACON_NAME "ESP32_Exhibit1"
6  void setup() {
7      BLEDevice::init(BEACON_NAME);
8      // Create the BLE Server
9      BLEServer *pServer = BLEDevice::createServer();
10     // Set up advertising
11     BLEAdvertising *pAdvertising = BLEDevice::getAdvertising();
12     pAdvertising->addServiceUUID(BLEUUID((uint16_t)0xFEAA)); // Eddystone (Optional)
13     pAdvertising->setScanResponse(true);
14     pAdvertising->setMinPreferred(0x06); // Optimization for discovery
15     // Start advertising the Name
16     BLEDevice::startAdvertising();
17 }
18 void loop() {
19 |

```

Figure 5: Firmware Implementation (C++/Arduino)

This configuration enables the ESP32 to function as a non connectable BLE advertiser. Each beacon is assigned a unique name according to the exhibit number. The Unity application filters these names and maps them to specific fossil exhibits. This approach improves wayfinding because users do not need to manually search for exhibits using static maps or printed directions. Instead, the application provides real-time proximity feedback and guides users toward the correct exhibit location.

AI Virtual Guide System

The AI Virtual Guide is one of the key innovative components of the proposed AR Fossil Tour system. It enables users to interact with museum content through voice based communication rather than only reading static descriptions. The guide is designed to provide exhibit explanations, answer user questions, generate quiz content, and support age-appropriate learning.

The system uses a **Retrieval Augmented Generation (RAG-lite)** approach to ensure that the AI provides accurate and controlled responses. Instead of allowing the AI model to generate answers freely, predefined factual exhibit data is first provided as the knowledge source. The AI then uses this verified content to generate explanations in a suitable tone and difficulty level. This reduces the risk of hallucination and ensures that the information remains scientifically accurate.

A special feature of the AI guide is its **age-based adaptation system**. Users are grouped into age categories such as **5–8**, **9–12**, and **13–17**. Based on the selected age group, the AI changes the complexity, vocabulary, and explanation style.

For younger children, the guide uses simple words, short sentences, and storytelling style explanations. For older users, it provides more detailed and scientifically accurate descriptions. This makes the learning experience more personalized and suitable for different cognitive levels.

The AI interaction pipeline consists of four main stages. First, the user’s voice input is captured and converted into text using Speech to Text technology such as Whisper. Next, the transcribed input is processed using a GPT based NLP model together with the relevant exhibit facts and age based prompt. Then, the AI generates a suitable response based on the predefined knowledge source. Finally, the response is converted into speech using Text to Speech technology and presented through the Virtual Guide.

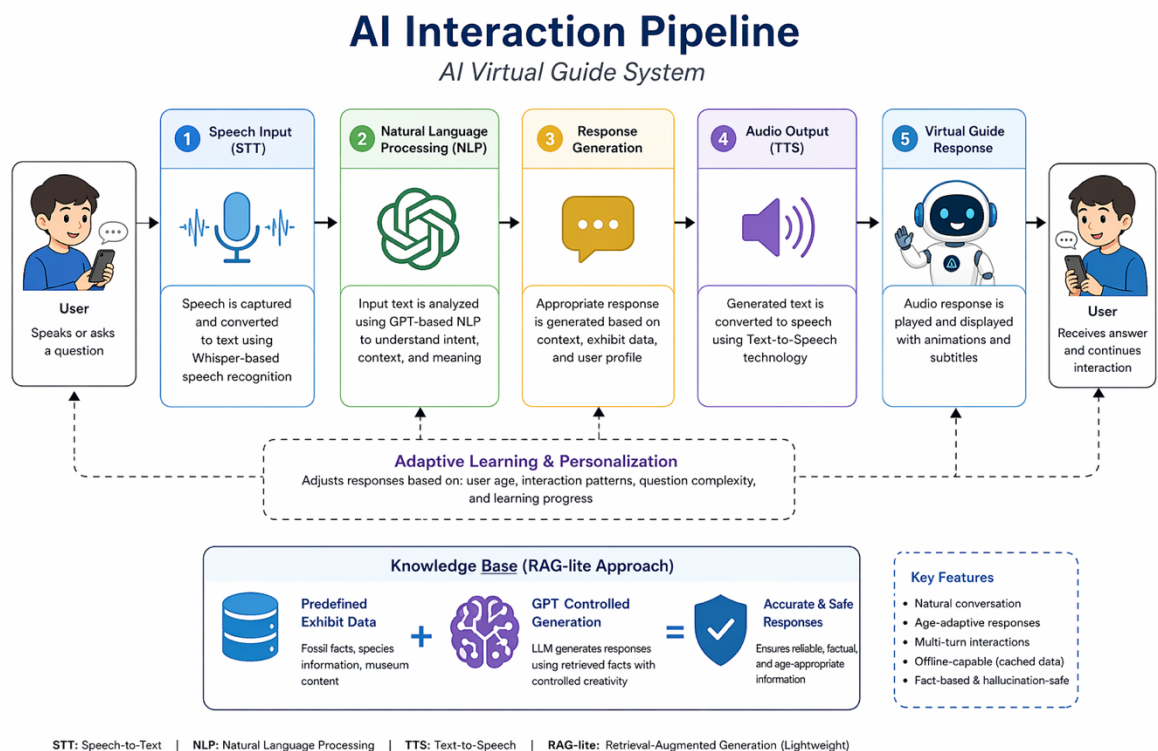


Figure 6: AI Virtual Guide Pipeline with RAG and Age Based Adaptation

Sample Code Snippet – RAG-Based AI Query Handling

```
string roleChunk = $"You are an educational AR guide for a  
'{ageGroup}' user.";  
  
string taskChunk =  
    $"Generate a 3-question multiple choice quiz about the exhibit:  
'{exhibitTopic}'.";  
  
string descriptionChunk = string.IsNullOrEmpty(exhibitDescription)  
    ? ""  
    : $"CONTENT SOURCE: Use these specific facts to create the  
questions: {exhibitDescription}\n";  
  
string ageGuidelinesChunk = GetAgeSpecificGuidelines(ageGroup,  
true);  
  
string contentRulesChunk =  
    "Create specific, factual questions about the animal's behavior,  
diet, unique body parts, or habitat. Do not ask generic questions.";  
  
string formatChunk =  
    "Generate the response strictly as a JSON object with 3  
questions, 4 answers, and correctAnswerIndex.";  
  
string systemPrompt =  
    roleChunk + "\n" +  
    taskChunk + "\n" +  
    descriptionChunk + "\n" +  
    ageGuidelinesChunk + "\n" +  
    contentRulesChunk + "\n" +  
    formatChunk;  
  
req.messages = new Message[]  
{  
    new Message { role = "system", content = systemPrompt },  
    new Message { role = "user", content = "Generate the quiz now." }  
};
```

This function sends the user’s question together with verified exhibit facts and the selected age group to the AI service. The `exhibitFacts` parameter acts as the RAG knowledge source, while the `ageGroup` parameter controls the explanation style. As a result, the Virtual Guide can provide accurate, personalized, and child-friendly explanations about each fossil exhibit.

Sample Code Snippet – Age-Based Prompt Selection

```
private string GetAgeSpecificGuidelines(string ageGroup, bool
isQuiz)
{
    if (ageGroup == "5-8")
    {
        return "Use very simple words. Keep sentences short. Make
the tone friendly and fun. Focus on basic facts such as food, size,
sound, and appearance.";
    }
    else if (ageGroup == "9-12")
    {
        return "Use clear and descriptive language. Ask medium-level
questions about animal behavior, adaptations, habitat, and
survival.";
    }
    else if (ageGroup == "13-17")
    {
        return "Use accurate scientific terms. Provide detailed
explanations and ask challenging questions about evolution,
paleontology, and ecological relationships.";
    }

    return "Use clear educational language suitable for general
users.";
}
```

A key feature of the AI guide is its use of controlled content chunks. Instead of allowing the AI model to generate answers freely, the system provides verified exhibit facts as a content source. This lightweight Retrieval-Augmented Generation (RAG) approach helps reduce hallucination and ensures that generated responses remain scientifically accurate and relevant to the selected exhibit.

Sample Code Snippet – Prompt Chunk Construction

```
string roleChunk = $"You are an educational AR guide for a
'{ageGroup}' user.";

string taskChunk =
    $"Generate a 3-question multiple choice quiz about the exhibit:
'{exhibitTopic}'.";

string descriptionChunk = string.IsNullOrEmpty(exhibitDescription)
    ? ""
    : $"CONTENT SOURCE: Use these specific facts to create the
questions: {exhibitDescription}\n";

string ageGuidelinesChunk = GetAgeSpecificGuidelines(ageGroup,
true);
```

```

string contentRulesChunk =
    "Create specific, factual questions about the animal's behavior,
    diet, unique body parts, or habitat. Do not ask generic questions.";

string formatChunk =
    "Generate the response strictly as a JSON object with 3
    questions, 4 answers, and correctAnswerIndex.";

string systemPrompt =
    roleChunk + "\n" +
    taskChunk + "\n" +
    descriptionChunk + "\n" +
    ageGuidelinesChunk + "\n" +
    contentRulesChunk + "\n" +
    formatChunk;

```

Gamification and Quiz Module

The Gamification and Quiz Module enhances user engagement by encouraging active participation during the museum tour. Instead of passive observation, users interact with exhibits through quiz-based learning activities.

The system generates multiple-choice questions based on exhibit content and adapts difficulty according to the user's age group.

Gamification is achieved through rewards and progression mechanisms. Users earn points for correct answers and unlock subsequent exhibits, creating a structured and motivating experience. Additionally, quiz interactions and performance data are logged for evaluating learning outcomes. This approach improves knowledge retention and makes the museum experience more interactive compared to traditional methods.

User Interface Design (Child-Friendly UI)

The system is specifically designed to support young users (ages 5–12), who represent the primary target audience. The user interface follows child-centered design principles to ensure accessibility, usability, and engagement.

Key design considerations include:

- **Large Touch Targets:** Buttons are designed with sizes greater than 48dp to ensure easy interaction for children.
- **Simple Navigation:** Minimal steps are required to perform actions such as scanning, interacting, and answering quizzes.
- **Bright and Engaging Colors:** The interface uses vibrant colors to attract attention and maintain interest.
- **Audio + Visual Support:** All information is presented using both voice and subtitles to support different learning styles.
- **Interactive Elements:** Features such as rotating 3D models and tapping the virtual guide increase engagement.



Figure 7: Onboarding Screens



Figure 8: Loading screen, Main menu and Map

Data Logging and Analytics

The system uses Firebase Realtime Database to collect interaction data.

Data Collected,

- Session duration
- Quiz scores
- Interaction frequency
- Navigation paths

Sample Code Snippet – Event Logging

```
public void LogEvent(string eventName)
{
    FirebaseManager.Log(eventName);
}
```

This enables real-time data collection for evaluating engagement and learning outcomes.

Commercialization Aspects of the Product

Introduction to Commercialization Strategy

The proposed AR Fossil Tour system is not only a research prototype but also a commercially viable digital heritage solution designed for deployment in museums, educational institutions, and tourism environments. The commercialization strategy focuses on delivering a low cost, scalable, and user centric platform that enhances visitor engagement while minimizing operational overhead for institutions.

Unlike traditional museum technologies that require expensive hardware installations (e.g., VR headsets, kiosks), this system leverages visitor owned smartphones combined with low cost BLE infrastructure, making it highly adaptable for developing countries such as Sri Lanka.

Value Proposition

The value proposition of the proposed system can be analyzed across three primary stakeholder groups: museums, visitors, and educational institutions, each benefiting from the integration of augmented reality (AR), artificial intelligence, and interactive learning technologies.

From the perspective of museums, the system enhances visitor engagement by increasing dwell time at exhibits through immersive and interactive content delivery. Previous studies have demonstrated that AR based museum applications significantly improve user engagement and overall visitor experience compared to traditional display methods [2], [4]. The integration of an AI driven Virtual Guide further reduces dependency on human guides, thereby improving operational efficiency. Additionally, the system enables digital transformation without requiring major infrastructural modifications, making it a cost effective and scalable solution, particularly for resource constrained environments [2].

For visitors, including students and tourists, the system provides an enriched learning experience characterized by interactivity and immersion. Research indicates that AR technologies improve learning motivation, engagement, and knowledge retention by combining visual, auditory, and interactive elements [3], [4]. The ability to explore exhibits in a personalized and self-paced manner supports diverse learning styles and enhances user autonomy. Furthermore, the inclusion of multilingual and accessible content improves inclusivity, ensuring that the system can effectively cater to a broader audience with varying educational backgrounds and language preferences.

In the context of educational institutions, the system serves as a supplementary learning tool that aligns with formal curricula while extending learning beyond traditional classroom settings. AR-based educational systems have been shown to facilitate experiential and inquiry-based learning, leading to improved conceptual understanding and academic performance [1][3]. By transforming museum visits into interactive learning experiences, the system enhances knowledge retention and student engagement. Moreover, it provides an affordable and practical solution for enriching educational field trips, supporting modern pedagogical approaches that emphasize active and student-centered learning.

Market Analysis

The proposed system is applicable across multiple domains, including museums, schools, tourism, and universities. Museums such as natural history, archaeological, and science institutions can use the system to enhance visitor engagement through interactive exhibits. Schools benefit by improving educational field trips with experiential learning. In tourism, the system supports heritage and eco-tourism by providing interactive content. Universities can use it for research and technology-based learning.

Segment	Description
Museums	Natural history, archaeological, science museums
Schools	Educational field trips
Tourism	Heritage and eco tourism
Universities	Research and learning environments

The market demand is driven by increasing interest in digital learning experiences and the growth of AR based educational tools. There is also a shift toward interactive museum environments, replacing traditional static displays.

In Sri Lanka, the demand is further supported by limited adoption of advanced museum technologies, high smartphone usage among youth, and the need for cost effective solutions. These factors make mobile AR a practical and scalable approach.

Product Model

The proposed system is designed as a hybrid software service product that integrates multiple technologies to deliver an interactive and scalable museum experience. The system is composed of the following core components:

Core Components

- Mobile AR application
- AI based Virtual Guide system
- BLE based navigation infrastructure
- Content management system

These components work together to provide real time visualization, interactive learning, and efficient navigation within the museum environment. The mobile AR application serves as the primary user interface, while the AI Virtual Guide enhances user interaction through personalized explanations. The BLE infrastructure enables proximity based navigation, and the content management system ensures efficient handling and updating of exhibit information.

The deployment model supports optional BLE beacons for navigation, allowing flexible implementation based on the museum environment. Additionally, the system includes offline content delivery to ensure reliability and uninterrupted user experience in low connectivity conditions.

Revenue Model

Several monetization strategies can be adopted to ensure the financial sustainability of the proposed system while maintaining accessibility for different user groups. These strategies are designed to accommodate institutional, educational, and public sector stakeholders.

Monetization Models

- **Licensing Model**

Museums pay an annual licensing fee, which includes system updates, maintenance, and technical support.

- **Subscription Model**

Schools and educational institutions subscribe to the system for academic use, with tier-based pricing options such as basic and premium content packages.

- **Freemium Model**

A basic AR experience is provided free of charge to attract users, while advanced features such as the AI Virtual Guide and interactive quizzes are offered as paid upgrades.

- **Government / NGO Funding**

Financial support can be obtained through cultural preservation initiatives and educational technology grants, particularly for public museums and learning programs.

These monetization approaches provide flexibility by supporting both revenue generation and wider adoption. They enable the system to be deployed in diverse contexts, including resource-constrained environments, while ensuring long term operational viability.

Cost Structure

The system is intentionally designed to minimize overall costs, making it suitable for deployment in resource constrained environments such as public museums. The cost structure is divided into development, operational, and infrastructure components, ensuring affordability while maintaining functionality.

Cost Structure

- **Development Costs**

Include software development using platforms such as Unity and AI integration, along with 3D modeling and animation required for AR visualization.

- **Operational Costs**

Consist of low cost BLE beacons (approximately \$3 – 8 per unit) and periodic maintenance and system updates.

- **Infrastructure Costs**

Remain minimal, as the system does not require dedicated servers due to its offline content delivery capability.

The system offers a significant cost advantage compared to traditional Virtual Reality (VR) based solutions. It eliminates the need for expensive hardware such as VR headsets, avoids installation costs, and does not require specialized training for users or staff. This makes the proposed solution highly scalable, practical, and economically viable for real world museum environments.

Scalability and Expansion

The proposed system is designed with high scalability, enabling flexible deployment and future expansion across different domains and use cases. This scalability is achieved through both deployment and content-level adaptability. From a deployment perspective, the system supports horizontal scaling, allowing it to be implemented across multiple museums and extended to various types of exhibits without requiring significant infrastructure changes. In terms of content scalability, the system enables the easy addition of new fossils, exhibits, and educational narratives through a modular content management approach. Furthermore, it supports localization features, including multi language content delivery, making the system accessible to diverse user groups. These capabilities ensure that the system can be continuously expanded and updated without extensive redevelopment, allowing institutions to adapt the platform based on evolving user needs.

Future Expansion Opportunities

- Multi user AR experiences for collaborative interaction
- Cloud based analytics dashboard for real time insights
- Integration with tourism applications

These future enhancements highlight the system's potential for long-term growth, enabling advanced features and broader ecosystem integration while maintaining its core functionality.

Competitive Advantage

The proposed system offers several unique advantages that distinguish it from existing museum technologies, particularly in terms of integration, affordability, and contextual relevance.

Key Advantages

- Combines AR, AI, and BLE based navigation into a unified system
- Focuses on low cost implementation suitable for resource constrained environments
- Supports inquiry based learning through interactive engagement
- Designed specifically for the Sri Lankan museum and educational context

These features collectively enhance the effectiveness of the system by providing an immersive, intelligent, and accessible solution.

Comparison with Existing Systems

Existing museum technologies are often limited in scope, typically focusing on a single functionality such as AR visualization, audio guides, or QR-based content delivery. Many AR-based systems provide only static overlays or predefined animations without incorporating intelligent interaction or navigation support [2] [6]. Similarly, QR code based systems require users to manually scan markers, which does not assist in the exhibit discovery or wayfinding process.

In contrast, the proposed system integrates AR visualization, AI driven interaction, and BLE based navigation into a single platform, enabling both exhibit discovery and interactive learning. Unlike traditional audio guides or linear content delivery systems, the AI Virtual Guide supports inquiry-based learning, allowing users to actively request information rather than passively receiving it. Research has shown that interactive and inquiry based AR systems significantly improve engagement and learning outcomes compared to passive methods [3] [4].

Furthermore, many existing solutions rely on expensive infrastructure or specialized hardware, such as VR systems, limiting their scalability and adoption. Studies indicate that such systems often require high setup costs and controlled environments, reducing their practicality in real world museum settings [2].

The proposed system addresses this limitation by utilizing low cost mobile based technology, making it more suitable for deployment in developing regions.

Additionally, most existing systems are designed for global contexts without localization. However, cultural heritage applications emphasize the importance of context-specific design for improving usability and accessibility [2], [7]. The proposed system is specifically tailored to the Sri Lankan context, considering local educational needs, technological constraints, and user behavior, thereby increasing its relevance and effectiveness.

Feature	Existing Systems	Proposed System
AR Visualization	Limited / Static	Interactive 3D models
AI Interaction	Not available	AI Virtual Guide
Navigation Support	QR/manual navigation	BLE based real time navigation
Learning Approach	Passive	Inquiry based
Cost	Medium to High	Low cost implementation
Localization	Generic/global	Sri Lanka specific design

Table 1: Comparison Between Existing Systems and Proposed AR System

Risk and Commercial Challenges

The proposed system, while offering significant advantages, may face several commercialization risks that could affect its adoption and long term sustainability. These risks are primarily related to user acceptance, technical performance, device compatibility, and content management. However, appropriate mitigation strategies have been identified to address each of these challenges effectively.

Risk	Mitigation Strategy
Low adoption	User friendly design
Technical issues	Offline capability
Device limitations	System optimization
Content maintenance	Content Management System (CMS) tools

Table 2: Commercialization Risks and Mitigation Strategies

Low adoption risk can be minimized by ensuring an intuitive and user-friendly interface that caters to diverse user groups. Technical issues, particularly those related to connectivity, are addressed through offline functionality, ensuring system reliability. Device limitations are mitigated through optimization techniques to support mid range smartphones. Finally, content maintenance challenges are managed the use of a content management system, allowing easy updates and scalability.

Sustainability

The proposed system is designed to ensure long term sustainability by minimizing resource consumption and supporting continuous updates. The use of digital content allows museums to update information without requiring physical modifications to exhibits, reducing material waste and operational effort. Additionally, the system utilizes low energy BLE devices, which consume minimal power and contribute to energy efficiency. The reusable software architecture further enhances sustainability by allowing the system to be adapted and extended across different museums and contexts without significant redevelopment. These factors collectively ensure that the system remains environmentally friendly, cost effective, and scalable over time.

Testing and Implementation

Introduction

The testing and implementation phase is a critical component of this study, as it validates the effectiveness of the proposed AR based museum system in a real world environment. This phase aims to evaluate whether the system provides measurable improvements over traditional museum methods. Specifically,

the evaluation focuses on three key aspects: wayfinding efficiency, user engagement, and learning outcomes. By comparing user performance in both traditional and AR based scenarios, the study assesses the practical impact of the system on visitor experience and educational effectiveness.

Implementation Environment

The system was deployed in a real museum setting to ensure realistic testing conditions. The implementation was conducted at the **Colombo National Museum, Sri Lanka**, which provides a suitable environment with diverse exhibits and typical visitor behavior.

A total of **30 students** participated in the study. The participants were selected based on specific criteria to ensure consistency and reliability of results. The selected participants belonged to the age group of **5 –18 years**,

had basic familiarity with smartphone usage and had no prior exposure to the developed AR system. This ensured that the results were not influenced by previous experience or bias.

The controlled selection of participants and the real-world testing environment contribute to the validity of the experimental results and ensure that the findings accurately reflect practical usage scenarios.

Experimental Setup

The experimental setup was designed to compare the effectiveness of the proposed AR based museum system with traditional printed label methods. Participants were divided into two groups: a control group and an experimental group. The control group used conventional printed labels to explore museum exhibits, while the experimental group used the AR Fossil Tour application. Both groups were assigned the same set of exhibits to ensure consistency and fairness in the evaluation.

Group	Method Used
Control Group	Printed Labels
Experimental Group	AR Fossil Tour

Table 3: Experimental Groups and Methods Used

Testing Procedure

The testing procedure was conducted in three main stages to ensure a structured and controlled evaluation.

Step 1: Introduction

Participants were initially briefed about the purpose of the study and provided with clear instructions on how to proceed. This ensured that all participants had a uniform understanding of the tasks and reduced potential bias during the experiment.

Step 2: Exploration Phase

During this phase, participants explored the assigned exhibits using their respective methods. The control group relied on printed labels and read textual descriptions provided at each exhibit. In contrast, the experimental group used the mobile AR application, which allowed them to navigate exhibits using BLE-based guidance, interact with 3D models, engage with the AI Virtual Guide, and complete interactive quizzes within the application. This phase focused on capturing differences in engagement, navigation behavior, and interaction levels between the two groups.

Step 3: Assessment Phase

After completing the exploration phase, all participants were required to complete a standardized assessment using a Google Form quiz. The quiz contained the same questions as those presented within the AR application, ensuring consistency in evaluation. This assessment was used to measure knowledge retention and compare learning outcomes between the two groups.

Data Collection Methods

The data required for evaluation was collected using multiple methods to ensure a comprehensive analysis of the system's performance.

1. Quiz Scores

Quiz results were collected using Google Forms and the AR application. These scores were used to compare learning outcomes between the traditional method and the AR based system.

2. System Logs

System generated logs were used to record user interactions, including time spent, interaction frequency, and navigation patterns. These logs provide quantitative data for analyzing user engagement and system usage.

3. Observations

Direct observations were conducted during the testing phase to assess user behavior and engagement levels. This qualitative data helps in understanding user interaction patterns and overall experience.

Evaluation Method

The primary evaluation method used in this study is based on the comparison of average quiz scores between the control group and the experimental group. This approach enables the measurement of differences in learning outcomes between traditional and AR based methods.

The average score was calculated using the following formula:

$$\text{Average Score} = \frac{\sum \text{Scores}}{n}$$

where n represents the total number of participants. The results were then compared between the AR group average and the traditional group average to determine the effectiveness of the proposed system.

In addition to learning outcomes, several performance indicators were used to evaluate engagement and navigation efficiency.

Metrics Calculation

Learning (Quiz Score)

Learning performance was measured using quiz scores obtained from participants. The improvement in learning was calculated as:

$$\text{Improvement (\%)} = \frac{(\text{AR Avg} - \text{Traditional Avg})}{\text{Traditional Avg}} \times 100$$

Engagement (Time Spent)

Engagement was measured using the time spent by users during the exploration phase:

$$\text{Average Time} = \frac{\sum \text{Time Spent}}{n}$$

Higher time values indicate greater user engagement.

Navigation Efficiency (Time to Locate Exhibits)

Navigation efficiency was evaluated based on the time taken to locate exhibits:

$$\text{Average Navigation Time} = \frac{\sum \text{Navigation Time}}{n}$$

Lower values indicate better navigation performance.

Metric	Description
Learning	Quiz score comparison
Engagement	Average time spent
Navigation	Time taken to locate exhibits

This evaluation method provides a quantitative basis for comparing the effectiveness of the AR system against traditional museum methods.

Data Analysis

The collected data was analyzed using basic statistical methods to evaluate the effectiveness of the proposed system. The analysis primarily focused on comparing the performance of the control group and the experimental group across key metrics.

Statistical Analysis

Mean comparison was used to evaluate differences in quiz scores, engagement time, and navigation efficiency between the two groups. The average values for each metric were calculated and compared to identify performance variations. Additionally, percentage improvement was calculated to quantify the extent of improvement achieved by the AR based system over the traditional method.

Expected Results

Based on prior research and system design, it is expected that the experimental (AR) group will demonstrate higher quiz scores, indicating improved learning outcomes. Furthermore, increased engagement is anticipated, reflected through longer interaction time and higher activity levels. In terms of navigation, the AR group is expected to locate exhibits more efficiently, resulting in reduced navigation time compared to the control group.

Validity and Reliability

- Internal Validity - Internal validity was maintained by ensuring that both groups were given the same content and questions. This ensures that any differences in results are due to the method used (AR vs traditional), not the content itself.
- External Validity - The experiment was conducted in a real museum environment. This makes the results more realistic and applicable to actual museum settings.
- Reliability - Reliability was ensured by using a standardized Google Form quiz for all participants. This guarantees consistency in data collection and evaluation.

Limitations of Testing

The testing phase had several limitations that may affect the generalization of the results.

The sample size was limited to 30 students, which may not fully represent all types of museum visitors. Additionally, the testing was conducted in a single museum, which limits the scope of the study and may not reflect conditions in other museum environments. Another limitation is device variability, as different smartphones may have variations in performance, which can influence the user experience and results.

RESULTS AND DISCUSSION

Introduction

This chapter presents the results obtained from the experimental evaluation of the proposed AR based museum tour system and discusses the findings in relation to the research objectives. The evaluation focuses on three key aspects: learning outcomes, user engagement, and wayfinding efficiency, comparing the AR based system with traditional printed label methods.

The analysis is based on data collected from quiz scores, system logs, and observational studies conducted at the Colombo National Museum. The results provide both quantitative and qualitative insights into the effectiveness of the proposed system.

Results

Learning Outcome Results

Learning performance was evaluated using quiz scores obtained from both the control group (printed labels) and the experimental group (AR system). The average scores were calculated and compared to assess knowledge retention.

- Traditional Method Average Score: 44.0
- AR System Average Score: 60.68

The results show a significant improvement in learning performance for participants using the AR system.

The results show a significant improvement in learning performance for participants using the AR system.

$$\text{Improvement (\%)} = \frac{(60.68 - 44)}{44} \times 100 \approx 38\%$$

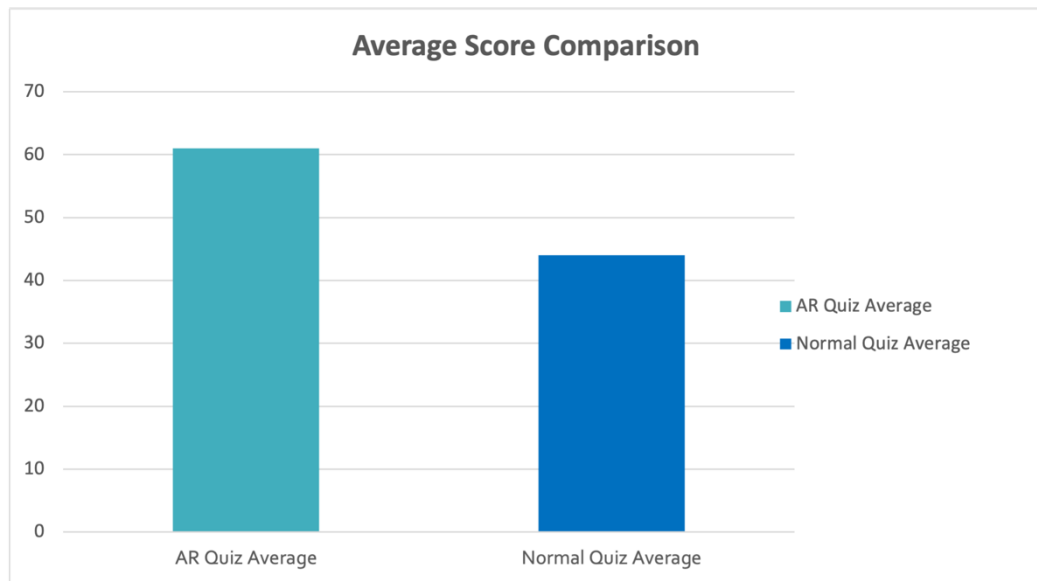


Figure 9: Average score comparison

Engagement Results

User engagement was measured using interaction logs and time spent during the exploration phase.

Key observations include:

- Over 600+ interaction events recorded (model rotations, zoom, quiz actions)
- Average engagement time significantly higher in AR group (~7 minutes per session)
- Traditional method engagement typically less than 1 minute per exhibit

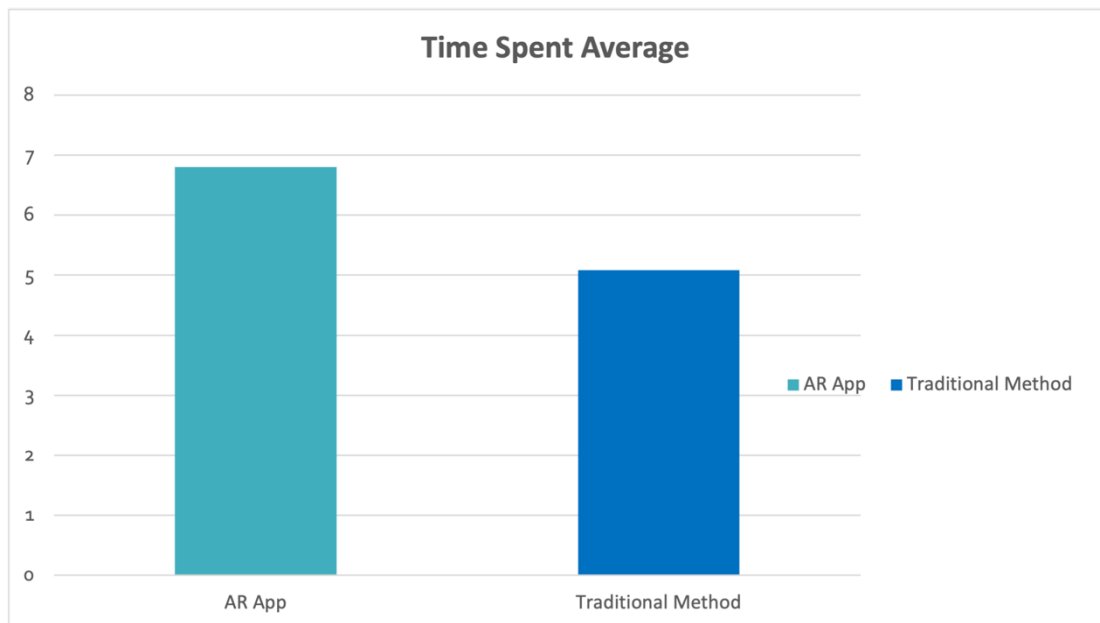


Figure 10: Average time spent across three exhibits

These results demonstrate that the AR system encourages active participation and prolonged interaction with exhibits.

Navigation Efficiency Results

Wayfinding efficiency was evaluated based on the time taken to locate exhibits and the navigation patterns observed.

Findings include:

- AR users located exhibits faster due to BLE-based guidance
- Reduced confusion and fewer wrong paths compared to control group
- Sequential navigation followed more consistently in AR group

The AR system significantly improved navigation efficiency within the museum environment.

Quiz Performance Analysis

Detailed quiz analysis revealed:

- **Average Correct Answers:** 2.82 / 3
- **Pass Rate:** Nearly 100% in AR group
- Faster completion time (~39 seconds per quiz)

Score(%)	Number of Users
0-25	0
26-50	11
51-75	9
76-100	10

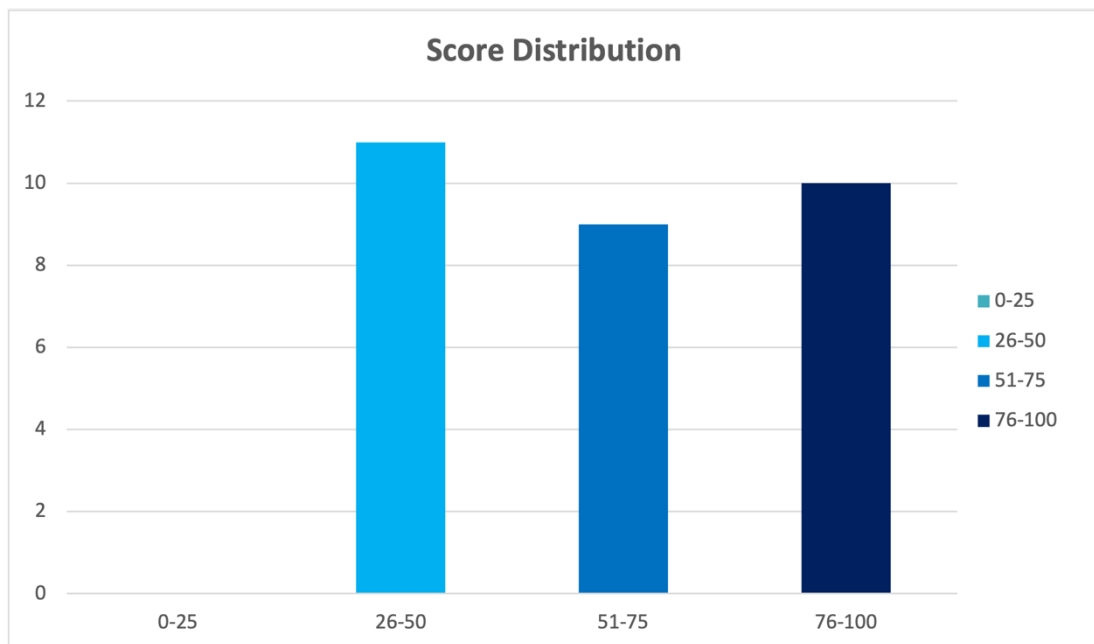


Figure 11: Quiz score distribution of participants

Interaction Behaviour Results

System logs revealed strong interaction patterns:

- Most interactions occurred with 3D models
- Elephant exhibit had highest engagement
- Drop off observed after second exhibit

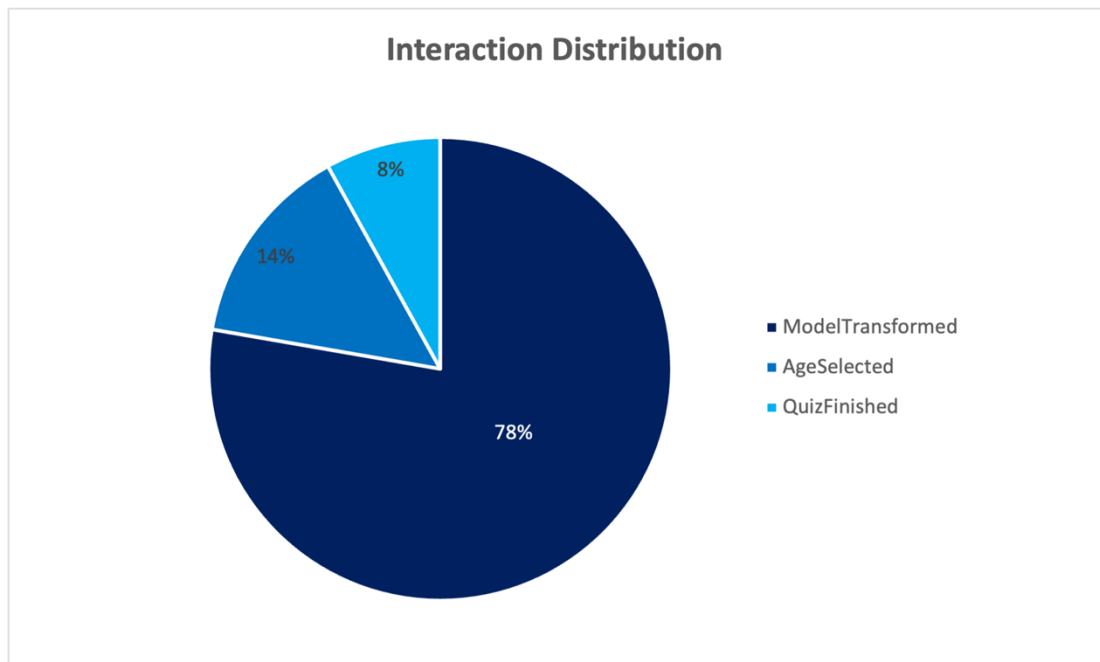


Figure 12: Interaction distribution of user events within the AR system

These results highlight both strengths and areas for improvement in content design.

Research Findings

Based on the experimental results and data analysis, several key findings were identified regarding the effectiveness of the proposed AR based museum system. The findings indicate that the AR system significantly improves learning outcomes compared to traditional printed label methods. The integration of 3D visualization, interactive quizzes, and AI driven guidance enhances users' cognitive understanding and supports better knowledge retention. This demonstrates that immersive and interactive learning environments are more effective than passive information delivery approaches.

In addition to learning improvements, the system showed a clear increase in user engagement. Participants exhibited higher interaction frequency, spent more time exploring exhibits, and actively engaged with the application through model interactions and quizzes. These results confirm that AR based environments provide a more engaging experience compared to static text based systems, encouraging users to participate actively rather than passively consuming information.

The study also revealed that BLE based navigation significantly enhances wayfinding efficiency within the museum environment. Users were able to locate exhibits more quickly and with less confusion, following a structured and guided exploration path. This finding validates the effectiveness of integrating proximity based navigation with AR systems to improve spatial awareness and movement within indoor environments. Furthermore, the inclusion of gamification elements, such as quizzes and reward mechanisms, played a crucial role in maintaining user motivation. These features encouraged users to continue exploring exhibits and reinforced learning through active participation and immediate feedback. Gamification contributed not only to increased engagement but also to improved retention of information.

Finally, the results indicate that the system is particularly effective for younger users, especially those in the 5–8 age group. The design of the content, interaction style, and visual elements aligns well with their cognitive abilities and learning preferences. This highlights the importance of age specific adaptation in AR based educational systems, ensuring that content remains accessible, engaging, and effective for the target audience.

Discussion

The results of this study demonstrate a clear distinction between traditional museum methods and the proposed AR based system in terms of learning effectiveness, engagement, and navigation efficiency. Traditional museum approaches primarily rely on static labels, which offer limited interaction and often result in passive learning experiences. In contrast, the AR based system introduces interactive 3D visualization,

real time guidance, and personalized learning support, enabling users to actively engage with exhibit content.

These findings align with previous studies indicating that traditional displays are less effective in maintaining visitor engagement compared to interactive technologies [2] [4]. The experimental results further confirm that the AR system outperforms traditional methods across all evaluated dimensions.

The integration of augmented reality significantly enhances the learning experience by transforming abstract or incomplete exhibit information into visually rich and interactive representations. By promoting active participation, the system allows users to explore content in a more meaningful and engaging manner. The combination of visual, audio, and interactive elements supports multimodal learning, which has been shown to improve knowledge retention and comprehension. Prior research highlights that AR based learning environments increase user motivation and academic performance [3][4], and the findings of this study are consistent with these conclusions, demonstrating a positive impact of AR on educational outcomes.

User behavior observed during the experimental phase further supports the effectiveness of the system. Participants exhibited higher levels of curiosity and excitement when interacting with AR features, leading to increased attention spans and more consistent exploration of exhibits. These observations are consistent with existing research on gamified AR systems, which report improved engagement and user satisfaction [7]. However, a decline in engagement was noted in later stages of the experience, suggesting potential content fatigue. This indicates the need for enhancements such as increased content variation, optimized session duration, and the introduction of additional motivational mechanisms to sustain user interest throughout the experience.

The system also demonstrated several practical strengths. The integration of BLE based navigation improved wayfinding efficiency by reducing user confusion and enabling structured exploration paths within the museum. Additionally, the system achieved high levels of engagement and improved learning outcomes while

maintaining a low cost implementation. This is particularly important for real world deployment, as previous studies emphasize the need for scalable and cost effective AR solutions in cultural heritage applications [2] [6].

The use of mobile devices further enhances accessibility, making the system suitable for a wide range of users without requiring specialized hardware.

Despite these advantages, certain limitations were identified during the study. The evaluation was conducted using a limited range of devices, which may affect the generalizability of the results across different hardware configurations. Furthermore, interaction patterns revealed that some exhibits attracted significantly more attention than others, indicating an imbalance in content design. Minor usability challenges were also observed, particularly among first time users. These limitations are consistent with challenges reported in previous AR research, where device compatibility and content optimization remain critical considerations [2].

Overall, the findings of this study highlight the strong potential of AR-based systems to modernize museum experiences. Mobile AR solutions provide a practical and scalable approach to enhancing visitor engagement without requiring significant infrastructure investment. The integration of interactive and immersive technologies contributes to improved learning outcomes and visitor satisfaction. This is particularly relevant for museums in Sri Lanka, where cost effective and innovative solutions are essential. Existing research in cultural heritage applications further supports the adoption of AR technologies as a means of improving accessibility, engagement, and educational value in museum environments [2] [4].

CONCLUSION AND FUTURE WORKS

Conclusion

This research aimed to evaluate whether Augmented Reality (AR)-based museum tours on visitors' mobile devices improve wayfinding, engagement, and learning outcomes compared to traditional printed-label tours. To achieve this, a mobile AR application integrating AR visualization, BLE-based navigation, AI interaction, and gamification was designed, developed, and tested in a real-world environment at the Colombo National Museum.

The findings of this study provide strong evidence that AR-based museum systems offer significant improvements over traditional methods. The results clearly demonstrate that participants who used the AR application achieved higher quiz scores compared to those who relied on printed labels, indicating improved knowledge retention. The calculated improvement in learning outcomes confirms that interactive and visual learning approaches enhance users' understanding of museum exhibits.

In terms of engagement, the AR system significantly increased user interaction and time spent exploring exhibits. The integration of 3D models, interactive elements, and gamification encouraged users to actively participate in the learning process rather than passively consuming information. Observations showed that users exhibited higher levels of curiosity, excitement, and attention when interacting with AR content. This highlights the effectiveness of immersive technologies in transforming user experience in informal learning environments.

The study also confirmed that AR-based navigation improves wayfinding efficiency. The BLE-based proximity system enabled users to locate exhibits more easily, reducing confusion and improving the overall flow of movement within the museum. Compared to traditional methods, where users often rely on maps or visual searching, the AR system provided a guided and structured exploration experience.

Another important contribution of this research is the development of a low cost, scalable solution tailored to the Sri Lankan context. Unlike many existing systems that require expensive infrastructure, the proposed system leverages widely available mobile devices and low cost BLE technology. This makes it highly suitable for deployment in developing regions where budget constraints limit the adoption of advanced technologies.

Furthermore, the integration of an AI based Virtual Guide introduced an inquiry-based learning approach, allowing users to actively seek information rather than passively receiving it. This aligns with modern educational theories that emphasize active learning and user centered interaction. The combination of AR, AI, and navigation technologies creates a comprehensive system that enhances both usability and educational value.

Despite its strengths, the study acknowledges certain limitations, including a relatively small sample size, testing in a single museum, and device variability. These factors may influence the generalizability of the results. However, the overall findings strongly support the research hypothesis and demonstrate the effectiveness of AR-based museum tours in improving visitor experience.

In conclusion, this research successfully validates that AR museum tours on visitors' phones significantly improve wayfinding, engagement, and learning outcomes compared to traditional printed label tours. The proposed system not only enhances educational effectiveness but also provides a practical and scalable solution for modernizing museum experiences.

Future Works

While the proposed system demonstrates significant potential, several opportunities exist for further improvement and expansion.

One key area for future work is the enhancement of **multi user AR experiences**. Enabling multiple users to interact within the same AR environment can support collaborative learning and improve social engagement within museum settings. This would allow groups of visitors, such as students, to explore exhibits together and share knowledge in real time.

Another potential improvement is the development of a **cloud-based analytics dashboard**. Such a system could provide museums with real time insights into visitor behavior, engagement patterns, and popular exhibits. This data can be used to optimize exhibit design, improve content delivery, and enhance overall visitor experience.

The integration of the system with **tourism applications and smart city platforms** is another promising direction. By connecting the AR system with external services, users can receive personalized recommendations, navigation assistance, and contextual information beyond the museum environment.

Future research can also focus on improving **content diversity and personalization**. Adaptive learning mechanisms can be introduced to tailor content based on user preferences, age groups, and learning levels. This would further enhance the effectiveness of the system for different types of users.

Additionally, expanding the system to support **multiple museums and cultural sites** would improve its scalability and impact. Localization features such as language support and culturally relevant content can be enhanced to cater to international audiences.

From a technical perspective, further optimization is needed to support a wider range of devices, including iOS platforms and newer Android versions. Improving AR tracking accuracy and BLE signal stability will also enhance system performance. Finally, future studies can involve larger and more diverse participant groups to improve the generalizability of results. Long-term studies can also be conducted to evaluate sustained learning outcomes and user engagement over time.

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APPENDICES